

Lab Sheet 2: DML Commands – INSERT, UPDATE and DELETE

Perform the following SQL tasks using suitable examples:

For the example take these tables

-- 1. Student Table

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    City VARCHAR(50),  
    Course_ID INT,  
    Age INT  
);
```

-- 2. Employee Table

```
CREATE TABLE Employee (  
    Emp_ID INT PRIMARY KEY,  
    Emp_Name VARCHAR(50),  
    Salary DECIMAL(10,2),  
    Dept_ID INT  
);
```

-- 3. Department Table

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

-- 4. Product Table

```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR(50),  
    Price DECIMAL(10,2),
```

Quantity INT

);

-- 5. Customer Table

CREATE TABLE Customer (

Customer_ID INT PRIMARY KEY,

Customer_Name VARCHAR(50),

City VARCHAR(50)

);

-- 6. Course Table

CREATE TABLE Course (

Course_ID INT PRIMARY KEY,

Course_Name VARCHAR(50),

Fees DECIMAL(10,2)

);

Questions :-

1. Insert one record into the Student table.

➔ INSERT INTO Student (Student_ID, Student_Name, City, Course_ID, Age)
VALUES (101, 'Amit Sharma', 'Delhi', 1, 20);

2. Insert five student records using separate INSERT statements.

➔ INSERT INTO Student VALUES (102, 'Rahul Verma', 'Meerut', 2, 21);

INSERT INTO Student VALUES (103, 'Priya Singh', 'Noida', 1, 19);

INSERT INTO Student VALUES (104, 'Anjali Gupta', 'Meerut', 3, 22);

INSERT INTO Student VALUES (105, 'Rohan Das', 'Mumbai', 2, 20);

INSERT INTO Student VALUES (106, 'Sneha Reddy', 'Bangalore', 1, 21);

3. Insert multiple records into the Student table using a single statement.

➔ INSERT INTO Student (Student_ID, Student_Name, City, Course_ID, Age) VALUES
(107, 'Vikram Malhotra', 'Delhi', 3, 23),
(108, 'Neha Kapoor', 'Meerut', 2, 20),
(109, 'Yash Vardhan', 'Pune', 1, 22);

4. Insert records into the Employee table.

→ INSERT INTO Employee (Emp_ID, Emp_Name, Salary, Dept_ID) VALUES
(501, 'Alice Smith', 45000.00, 10),
(502, 'Bob Johnson', 28000.00, 20),
(503, 'Charlie Brown', 60000.00, 10);

5. Insert records into the Department table.
→ INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'IT'),
(20, 'HR'),
(30, 'Finance');

6. Insert records into the Product table.
→ INSERT INTO Product (Product_ID, Product_Name, Price, Quantity) VALUES
(901, 'Laptop', 50000.00, 15),
(902, 'Mouse', 500.00, 0),
(903, 'Keyboard', 1200.00, 5);

7. Insert records into the Customer table.
→ INSERT INTO Customer (Customer_ID, Customer_Name, City) VALUES
(301, 'John Doe', 'New York'),
(302, 'Jane Doe', 'London');

8. Insert records into the Course table.
→ INSERT INTO Course (Course_ID, Course_Name, Fees) VALUES
(1, 'Computer Science', 80000.00),
(2, 'Data Science', 95000.00),
(3, 'Web Development', 60000.00);

9. Insert a student record with values for all columns.
→ INSERT INTO Student VALUES (110, 'Karan Johar', 'Mumbai', 3, 24);

10. Insert a student record by specifying only selected columns.
→ -- Omitting 'Age' and 'Course_ID' (they will default to NULL if no constraints exist)
INSERT INTO Student (Student_ID, Student_Name, City)
VALUES (111, 'Meera Bai', 'Jaipur');

11. Update the city of a particular student.
→ UPDATE Student
SET City = 'Kolkata'
WHERE Student_ID = 101;

12. Update the salary of a particular employee.
→ UPDATE Employee
SET Salary = Salary * 1.10;

13. Increase the salary of all employees by 10%.

sql

UPDATE Employee

```
SET Salary = Salary * 1.10;
```

14. Increase the price of all products by 5%.

sql

```
UPDATE Product
SET Price = Price * 1.05;
```

15. Change the course of a particular student.

sql

```
UPDATE Student
SET Course_ID = 2
WHERE Student_ID = 103;
```

16. Update the department of an employee.

sql

```
UPDATE Employee
SET Dept_ID = 30
WHERE Emp_ID = 502;
```

17. Update the city of all students from Meerut to Delhi.

sql

```
UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
```

18. Update the fees of a particular course.

sql

```
UPDATE Course
SET Fees = 85000.00
WHERE Course_ID = 1;
```

19. Update multiple columns of a student simultaneously.

sql

```
UPDATE Student
SET City = 'Chennai', Age = 22
WHERE Student_ID = 105;
```

20. Delete a particular student record using Student_ID.

sql

```
DELETE FROM Student
WHERE Student_ID = 110;
```

21. Delete all students belonging to a particular city.

sql

```
DELETE FROM Student
WHERE City = 'Pune';
```

22. Delete employees whose salary is below a specified amount.

sql

```
DELETE FROM Employee
WHERE Salary < 35000.00;
```

23. Delete products whose quantity is zero.

sql

```
DELETE FROM Product
WHERE Quantity = 0;
```

24. Delete records satisfying multiple conditions using AND/OR.

sql

```
DELETE FROM Student
WHERE Age > 22 AND City = 'Delhi';
```

25. Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

sql

```
-- Step A: Insert records to build initial state
INSERT INTO Student (Student_ID, Student_Name, City, Course_ID, Age)
VALUES
(201, 'Ravi Kumar', 'Patna', 1, 20),
(202, 'Suman Das', 'Ranchi', 2, 21);

-- Display table after INSERT execution
SELECT * FROM Student;
```

OUTPUT

1. Insert one record into the Student table.

Output: Query OK, 1 row affected

Student Table State:

Student_ID	Student_Name	City	Course_ID	Age
101	Amit Sharma	Delhi	1	20

2. Insert five student records using separate INSERT statements.

Output: Query OK, 5 rows affected (total)

Student Table State:

Student_ID	Student_Name	City	Course_ID	Age
101	Amit Sharma	Delhi	1	20
102	Rahul Verma	Meerut	2	21
103	Priya Singh	Noida	1	19
104	Anjali Gupta	Meerut	3	22
105	Rohan Das	Mumbai	2	20
106	Sneha Reddy	Bangalore	1	21

3. Insert multiple records into the Student table using a single statement.

Output: Query OK, 3 rows affected

Student Table State:

Student_ID	Student_Name	City	Course_ID	Age
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...	...	...	...	...
106	Sneha Reddy	Bangalore	1	21
107	Vikram Malhotra	Delhi	3	23
108	Neha Kapoor	Meerut	2	20
109	Yash Vardhan	Pune	1	22

4. Insert records into the Employee table.

Output: Query OK, 3 rows affected

Employee Table State:

Emp_ID	Emp_Name	Salary	Dept_ID
501	Alice Smith	45000.00	10
502	Bob Johnson	28000.00	20
503	Charlie Brown	60000.00	10

5. Insert records into the Department table.

Output: Query OK, 3 rows affected

Department Table State:

Dept_ID	Dept_Name
10	IT
20	HR
30	Finance

6. Insert records into the Product table.

Output: Query OK, 3 rows affected

Product Table State:

Product_ID	Product_Name	Price	Quantity
901	Laptop	50000.00	15
902	Mouse	500.00	0
903	Keyboard	1200.00	5

7. Insert records into the Customer table.

Output: Query OK, 2 rows affected

Customer Table State:

Customer_ID	Customer_Name	City
301	John Doe	New York
302	Jane Doe	London

8. Insert records into the Course table.

Output: Query OK, 3 rows affected

Course Table State:

Course_ID	Course_Name	Fees
1	Computer Science	80000.00
2	Data Science	95000.00
3	Web Development	60000.00

9. Insert a student record with values for all columns.

Output: Query OK, 1 row affected

Student Table State:

Student_ID	Student_Name	City	Course_ID	Age
...	...	...	...	...
109	Yash Vardhan	Pune	1	22
110	Karan Johar	Mumbai	3	24

10. Insert a student record by specifying only selected columns.

Output: Query OK, 1 row affected

Student Table State:

Student_ID	Student_Name	City	Course_ID	Age
110	Karan Johar	Mumbai	3	24
111	Meera Bai	Jaipur	NULL	NULL

11. Update the city of a particular student.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Student Table State Change:

- Student 101 changes from 'Delhi' to 'Kolkata'.

12. Update the salary of a particular employee.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Employee Table State Change:

- Employee 501 changes salary from 45000.00 to **48000.00**.

13. Increase the salary of all employees by 10%.

Output: Query OK, 3 rows affected Rows matched: 3 Changed: 3

Employee Table State:

Emp_ID	Emp_Name	Salary	Dept_ID
501	Alice Smith	52800.00 (was 48000)	10
502	Bob Johnson	30800.00 (was 28000)	20
503	Charlie Brown	66000.00 (was 60000)	10

14. Increase the price of all products by 5%.

Output: Query OK, 3 rows affected Rows matched: 3 Changed: 3

Product Table State:

Product_ID	Product_Name	Price	Quantity
901	Laptop	52500.00 (was 50000)	15
902	Mouse	525.00 (was 500)	0
903	Keyboard	1260.00 (was 1200)	5

15. Change the course of a particular student.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Student Table State Change:

- Student 103 (Priya Singh) updates Course_ID from 1 to 2.
-

16. Update the department of an employee.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Employee Table State Change:

- Employee 502 (Bob Johnson) updates Dept_ID from 20 to 30.
-

17. Update the city of all students from Meerut to Delhi.

Output: Query OK, 3 rows affected Rows matched: 3 Changed: 3

Student Table State Change:

- Students 102 (Rahul), 104 (Anjali), and 108 (Neha) change their city fields from 'Meerut' to 'Delhi'.
-

18. Update the fees of a particular course.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Course Table State Change:

- Course 1 (Computer Science) updates Fees from 80000.00 to 85000.00.
-

19. Update multiple columns of a student simultaneously.

Output: Query OK, 1 row affected Rows matched: 1 Changed: 1

Student Table State Change:

- Student 105 (Rohan Das) changes city to 'Chennai' and age to 22.
-

20. Delete a particular student record using Student_ID.

Output: Query OK, 1 row affected

Student Table State Change:

- Row for Student 110 (Karan Johar) is removed completely from the table.

21. Delete all students belonging to a particular city.

Output: Query OK, 1 row affected

Student Table State Change:

- Row for Student 109 (Yash Vardhan, who belongs to 'Pune') is removed.

22. Delete employees whose salary is below a specified amount.

Output: Query OK, 1 row affected

Employee Table State Change:

- Employee 502 (Bob Johnson, salary 30800.00) is removed because it is less than 35000.00.

23. Delete products whose quantity is zero.

Output: Query OK, 1 row affected

Product Table State Change:

- Product 902 (Mouse, quantity 0) is removed completely from the table.

24. Delete records satisfying multiple conditions using AND/OR.

Output: Query OK, 1 row affected

Student Table State Change:

- Student 107 (Vikram Malhotra: Age 23 and City 'Delhi') matches both criteria and is removed.
-

25. Sequential Pipeline Evaluation Output

Executing the sequential block yields three distinct `SELECT` outputs:

Output 1 (After `INSERT` statements execution)

Student_ID	Student_Name	City	Course_ID	Age
201	Ravi Kumar	Patna	1	20
202	Suman Das	Ranchi	2	21

Output 2 (After running the `UPDATE` query)

Student_ID	Student_Name	City	Course_ID	Age
201	Ravi Kumar	Gaya	1	20
202	Suman Das	Ranchi	2	21

Output 3 (After running the `DELETE` query)

Student_ID	Student_Name	City	Course_ID	Age
201	Ravi Kumar	Gaya	1	20